Video 2: Model and Validation Walkthrough

Oladapo Olaniyan | FIN-43900 Project 1 | Recording target: 4:00-4:30

Use this on your phone while recording. Read one block at a time and follow the green SHOW or SWITCH cue. Show only the project files on your computer.

Before you start

Order	Show on your computer	What to say
1	01-model.py	Model flow: forecast, FCFF, DCF, and bridge
2	02-assumptions.json	Key assumptions and their role
3	03-visible-output.html	Results and value per share
4	04-validation-checks.csv	Validation results
5	05-cold-run-evidence.json	Clean-environment rerun

Script

SHOW 1 - 01-model.py

Hi, I am Oladapo Olaniyan. This video explains how my PepsiCo valuation model works and how I checked it. The model moves from filing inputs, to assumptions, to forecast statements, free cash flow, discounted cash flow value, and finally a value per share.

STAY ON 01-model.py

The valuation uses free cash flow to the firm, or FCFF. FCFF focuses on cash generated by operations before financing decisions. Starting with operating profit, I apply taxes, noncash items, capital spending, and working-capital investment. I keep interest and other financing effects out of FCFF so the cash flow matches the enterprise-value approach.

SWITCH 2 - 02-assumptions.json

These are the key assumptions. They include revenue growth, margins, capital spending, taxes, WACC, terminal growth, and terminal ROIC. The model uses a 7.66 percent WACC and 2.5 percent terminal growth in the base case. These are model inputs, not claims that the future is certain.

SWITCH 3 - 03-visible-output.html

Here is the executed output. The base value is $117.29 per diluted-proxy share, compared with a recorded market price of $125.65. The low, base, and high outcomes are $76.12, $117.29, and $170.22. They are scenario outcomes, not probability-weighted price targets.

STAY ON 03-visible-output.html

The terminal value is 80.1 percent of enterprise value. That means long-run assumptions have a large effect on the result. This is why I avoid treating one DCF number as enough evidence to buy the stock immediately.

SWITCH 4 - 04-validation-checks.csv

These checks test whether the model works as documented. The run has 42 passes, zero failures, five open items, and two qualified items. The checks cover statement connections, cash flow, the valuation bridge, sensitivity direction, and known-answer cases. Passing these checks shows the calculations run; it does not prove every forecast assumption is correct.

SWITCH 5 - 05-cold-run-evidence.json

Finally, I ran the model in a clean environment. The rerun reproduced the principal output files. That shows the project can be rerun without relying on hidden local files. My main limitation is still the economics behind the assumptions, especially the current-date bridge, peer comparability, and long-run profitability.

After recording

Watch the recording once. Make the transcript from what you actually said, then correct only material finance terms, names, and numbers.